

Massive MIMO for 5G: How Big Can it Get?

Dr. Dhiraj Kumar Patel

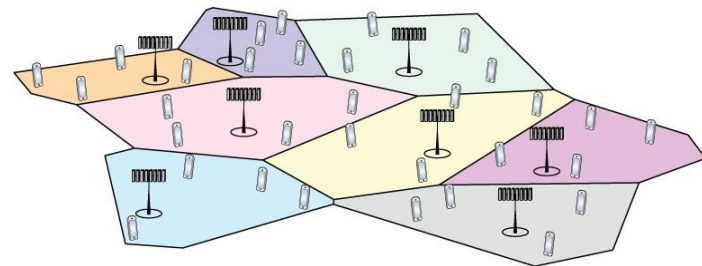
Outline

1. What is the Point of Massive MIMO?
2. How Big is Massive MIMO?
3. How Big Can it Get?

Improving Cellular Communications

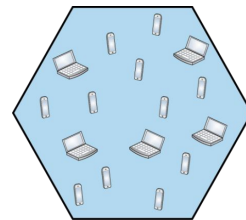
Cellular networks

Coverage area divided into cells
Users served by a base station



- Increase Network Throughput [bit/s/km²]

- Consider a given area



- Simple Formula for Network Throughput:

$$\underbrace{\text{Throughput}}_{\text{efficiency}} = \underbrace{\text{Cell density}}_{\text{bit/s/km}^2} \times \underbrace{8}_{\text{Cell/km}^2} \times \underbrace{\text{Available spectrum}}_{\text{Hz}} \times \underbrace{8}_{\text{bit/s/Hz/Cell}} \times \underbrace{\text{Spectral}}_{\text{bit/s/Hz/Cell}}$$

Conventional approach:

*Expensive to deploy
more base stations*

Spectrum is scarce:

*Plenty of mm-wave
spectrum, but only for
short-range scenarios*

Can we improve it?

*Improve technology,
upgrade base stations*

My focus:

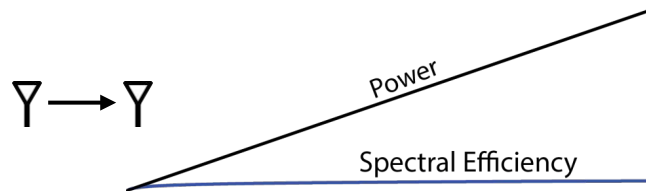
Spectrum <5 GHz

How to Increase Spectral Efficiency?

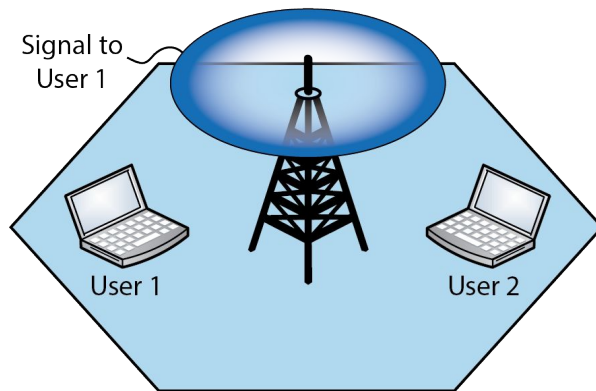
- Point-to-Point Spectral Efficiency:
 - Shannon's capacity limit:

$$\log_2 \left(1 + \frac{\text{Received Signal Power}}{\text{Interference Power} + \text{Noise Power}} \right) \quad [\text{bit/s/Hz/user}]$$

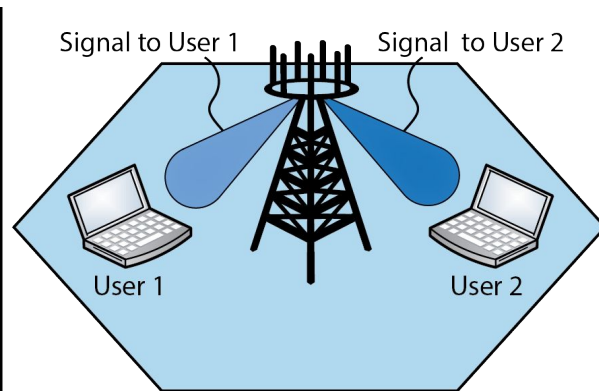
- Issue: 4 bit/s/Hz $\rightarrow$ 8 bit/s/Hz requires 17 $\times$ more power!



**Many Parallel
Transmissions:**
Spatially focused to
respective user

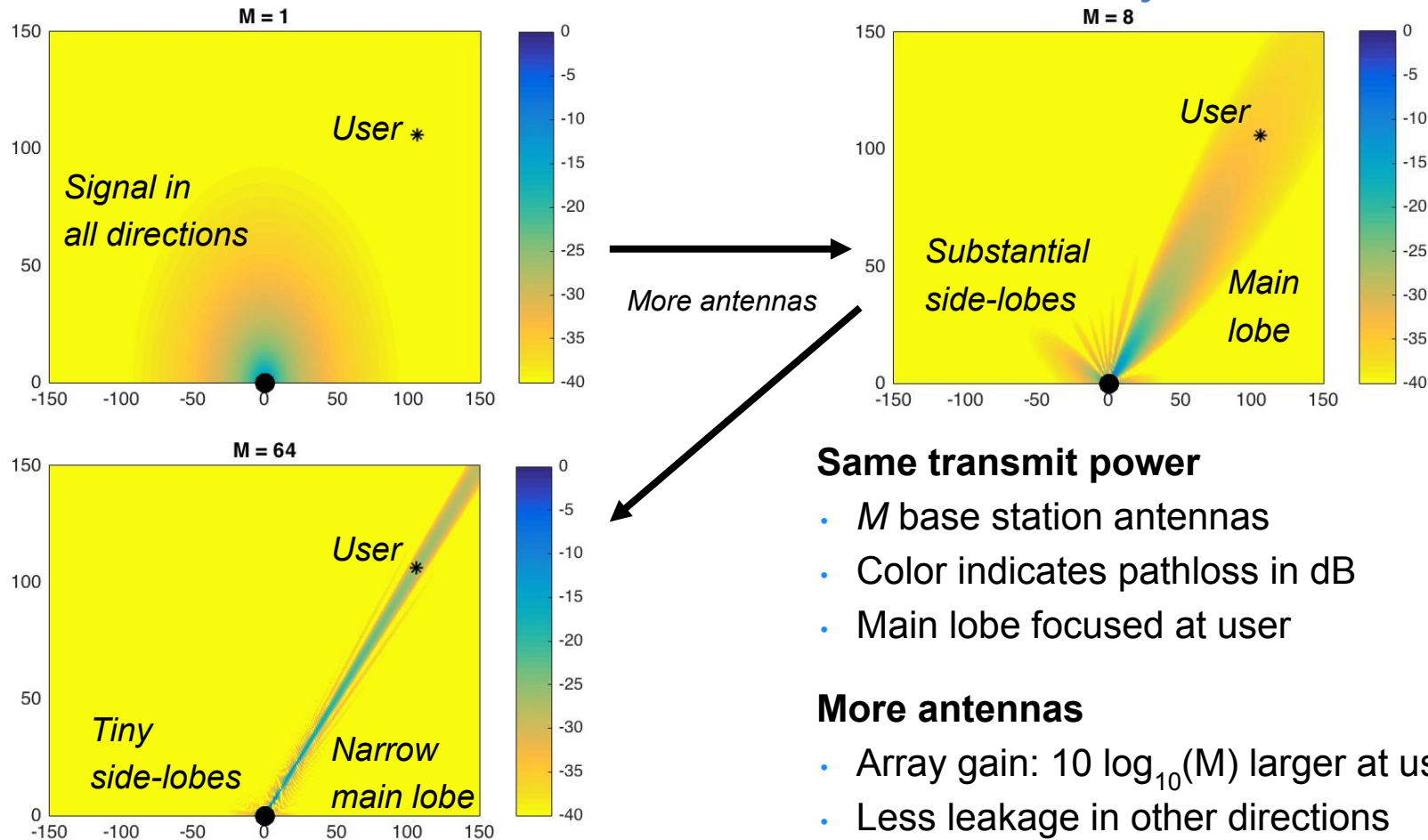


Single-Antenna Transmission



Multi-Antenna Transmission

More Antennas → More Directivity



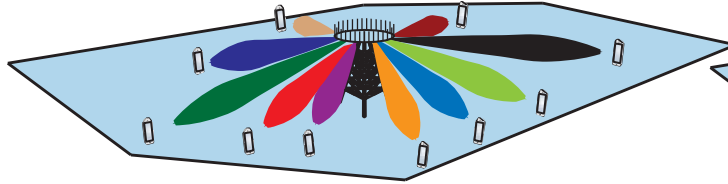
Same transmit power

- M base station antennas
- Color indicates pathloss in dB
- Main lobe focused at user

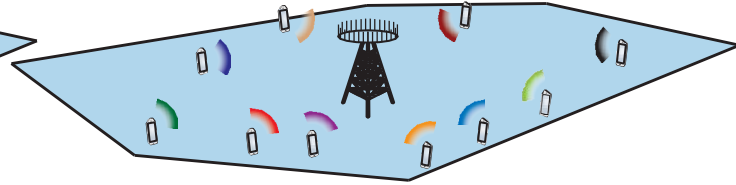
More antennas

- Array gain: $10 \log_{10}(M)$ larger at user
- Less leakage in other directions

Multi-User MIMO (Multiple-input Multiple-output)



Downlink: Multi-user beamforming



Uplink: Multi-user detection

Early Works

1987: J. Winters, “*Optimum combining for indoor radio systems with multiple users*,” IEEE Trans. Commun.

1990: S. Swales, M. Beach, D. Edwards, J. McGeehan, “*The performance enhancement of multibeam adaptive base-station antennas for cellular land mobile radio systems*,” IEEE Trans. Veh. Technol.

1991: S. Anderson, M. Millnert, M. Viberg, B. Wahlberg, “*An adaptive array for mobile communication systems*,” IEEE Trans. Veh. Technol.

Standards

2005: HC-SDMA

2008: LTE

2013: 802.11ac (WiFi)

Disappointing Results

Few antennas: Small theoretical gains

Insufficient channel knowledge:

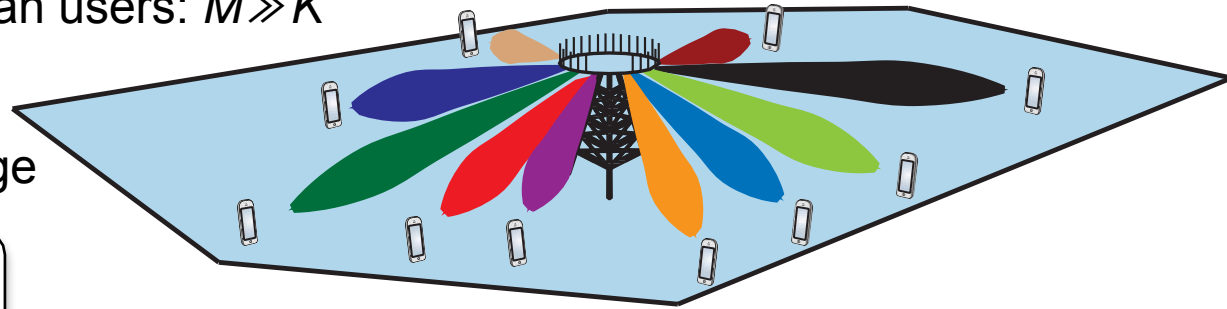
A lot of interference

Cell densification: Few users to multiplex

Massive MIMO: Taking Multi-User MIMO to the Next Level

- Main Characteristics
 - Many BS antennas; e.g., $M=200$ antennas, $K=40$ single-antenna users
 - Many more antennas than users: $M \gg K$
 - Very directive signals
 - Little interference leakage

*Spectral efficiency grows
with number of users!*



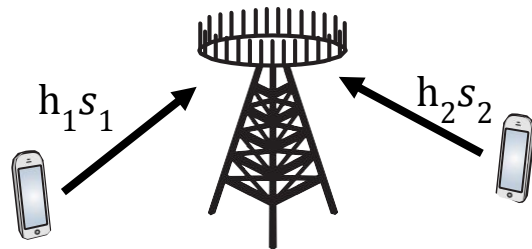
Seminal work: T. Marzetta, “Noncooperative Cellular Wireless with Unlimited Numbers of Base Station Antennas,” IEEE Trans. Wireless Communications, 2010.

- 2013 IEEE Guglielmo Marconi Prize Paper Award
- 2015 IEEE W. R. G. Baker Award

Motivated by Asymptotic Results

- Example: Uplink with i.i.d. Rayleigh Fading**

- Two users, send signals s_k for $k = 1, 2$
- Channels: $\mathbf{h}_k = [h_{k1} \dots h_{kM}]^T \sim \mathcal{CN}(0, \mathbf{I}_M)$
- Noise: $\mathbf{n} \sim \mathcal{CN}(0, \mathbf{I}_M)$
- Received: $\mathbf{y} = \mathbf{h}_1 s_1 + \mathbf{h}_2 s_2 + \mathbf{n}$



Linear Detector $\mathbf{v}_1$ for User 1: $\tilde{y}_1 = \mathbf{v}_1^H \mathbf{y} = \mathbf{v}_1^H \mathbf{h}_1 s_1 + \mathbf{v}_1^H \mathbf{h}_2 s_2 + \mathbf{v}_1^H \mathbf{n}$

Maximum ratio filter: $\mathbf{v}_1 = \frac{1}{\|\mathbf{h}_1\|} \mathbf{h}_1$

Signal remains: $\mathbf{v}_1^H \mathbf{h}_1 = \frac{1}{\|\mathbf{h}_1\|} \mathbf{h}_1^H \mathbf{h}_1 \xrightarrow{M \rightarrow \infty} \mathbb{E}[\|\mathbf{h}_1\|^2] = 1$

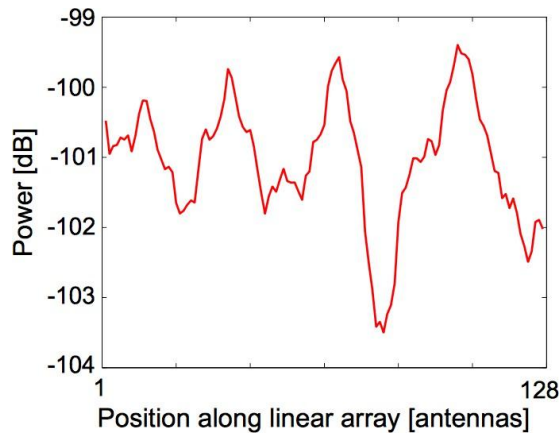
Interference vanishes: $\mathbf{v}_1^H \mathbf{h}_2 = \frac{1}{\|\mathbf{h}_1\|} \mathbf{h}_1^H \mathbf{h}_2 \xrightarrow{M \rightarrow \infty} \mathbb{E}[\mathbf{h}_1^H \mathbf{h}_2] = 0$

Noise vanishes: $\mathbf{v}_1^H \mathbf{n} = \frac{1}{\|\mathbf{h}_1\|} \mathbf{h}_1^H \mathbf{n} \xrightarrow{M \rightarrow \infty} \mathbb{E}[\mathbf{h}_1^H \mathbf{n}] = 0$

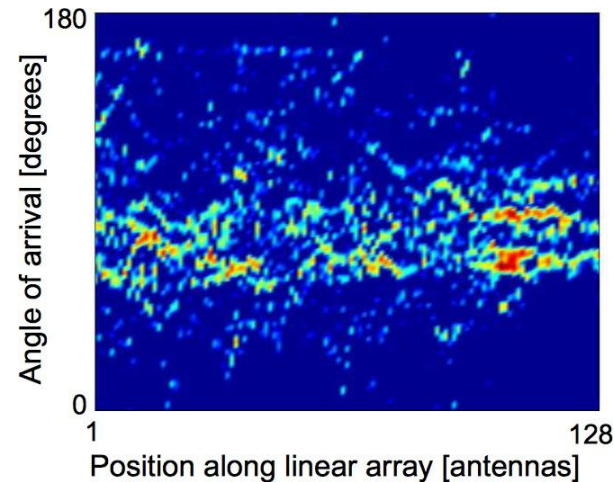
Asymptotically noise and interference free communication:

$$\tilde{y}_1 \xrightarrow{M \rightarrow \infty} s_1$$

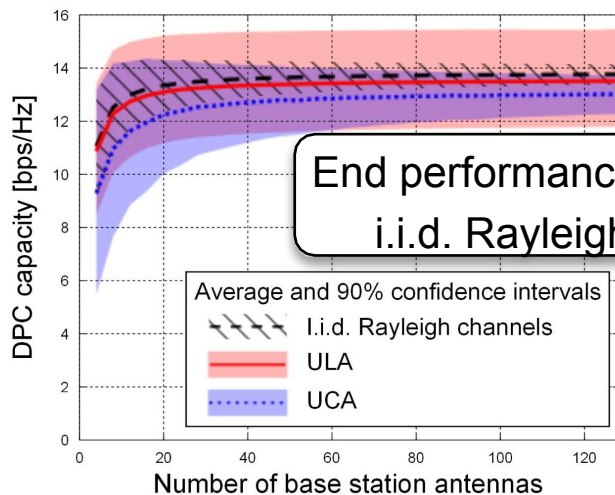
Practical Channels versus i.i.d. Rayleigh fading



Power variations over array



Each user has unique spatial signature



End performance similar to i.i.d. Rayleigh fading

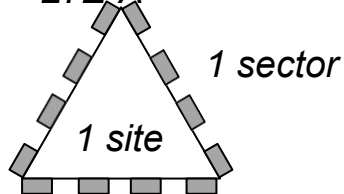
X. Gao, O. Edfors, F. Tufvesson, E. G. Larsson, "Massive MIMO in Real Propagation Environments: Do All Antennas

Contribute Equally?." Trans. on Comm., 2015

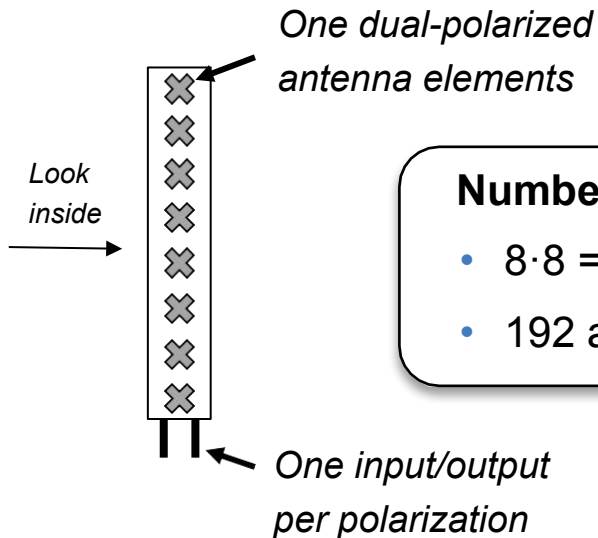
How Large is “Massive”?



3 sectors, 8-antenna
LTE-A



One dual-polarized
antenna panel



Number of Antennas?

- $8 \cdot 8 = 64$ per sector
- 192 antennas per site

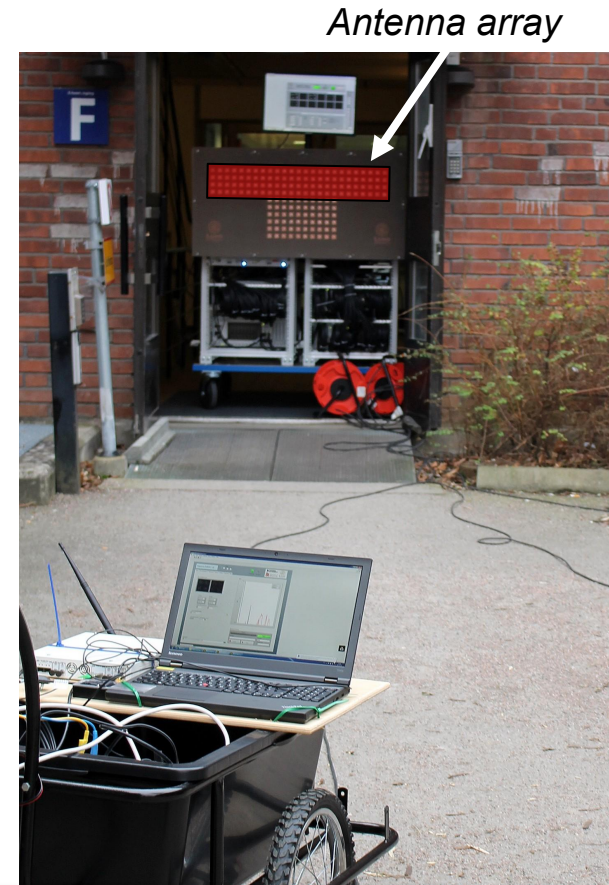
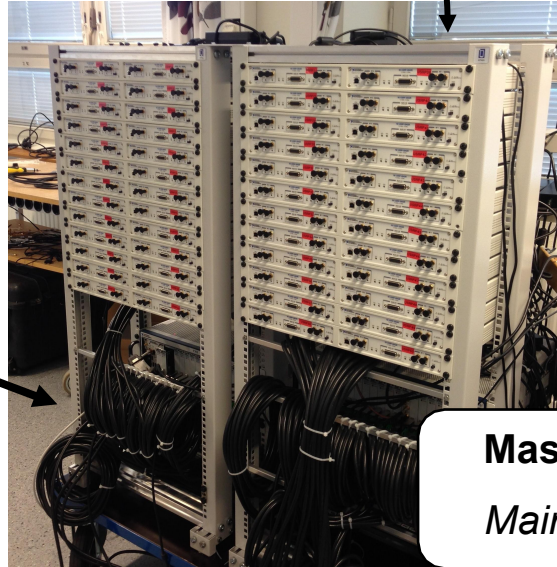
First Generation of Massive MIMO

Each antenna element becomes digitally steerable
Massive in numbers: 64 inputs/outputs instead of 8
Not massive in size: Same size 8-antenna LTE-A

Building Massive MIMO

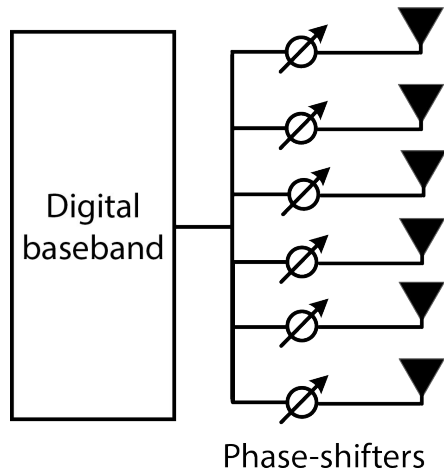
- Example: Lund University
 - Off-the-shelf hardware
 - 50 software-defined radios (SDRs) for 100 base station antennas
 - 6 SDRs for 12 users
 - 4 FPGAs for processing

A lot of cables...

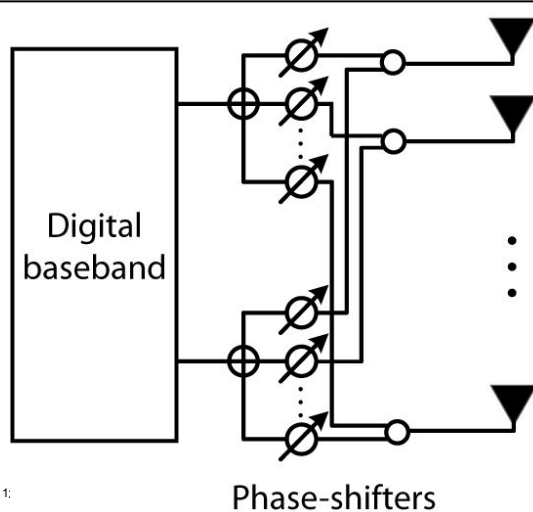
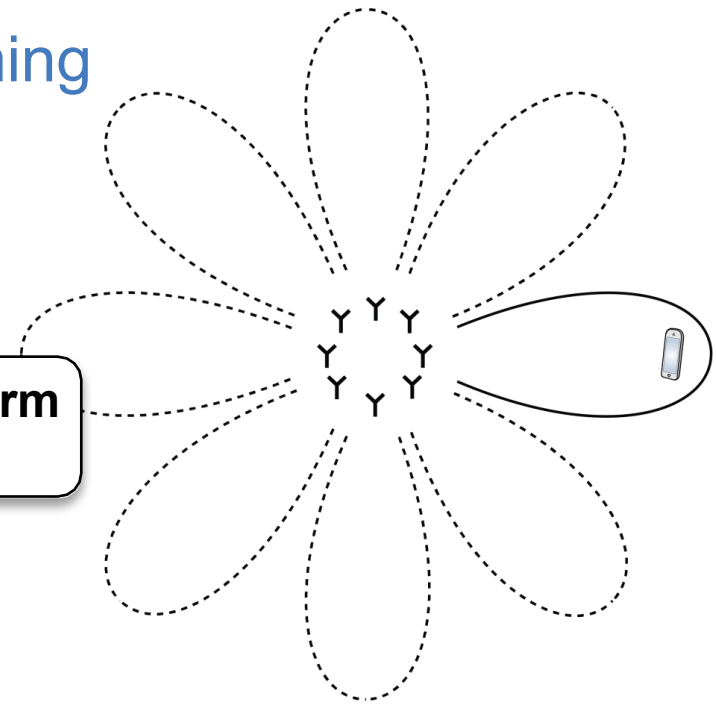


Massive MIMO uses mature hardware
Mainly an engineering effort to deploy it!

Analog Beamforming



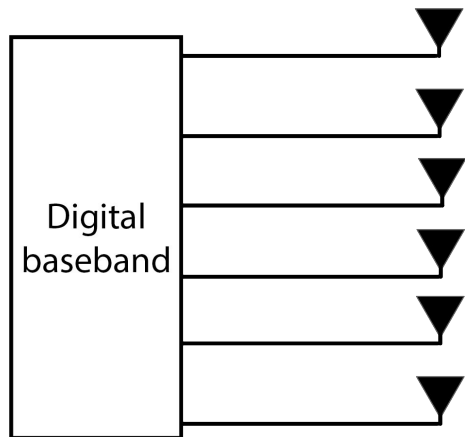
**Adjust phase-shifts to form
one angular beam**



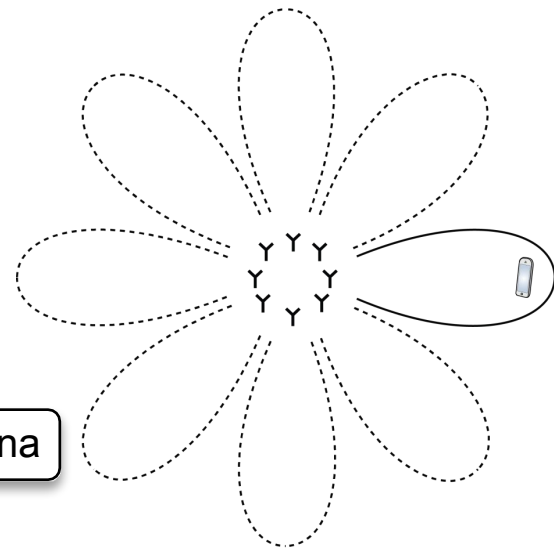
Hybrid analog-digital beamforming

- Create multiple angular beams
- Only one phase per signal and same power on all antennas

Digital Beamforming



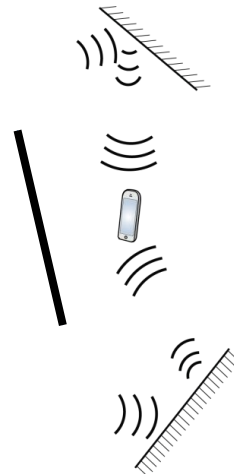
Full flexibility: Any signal on any antenna



Mandatory for non-line-of-sight scenarios

- Multipath propagation is complicated
- Require different phase and power per antenna and user

5G must have this flexibility! (<5 GHz)



“Channel State Information isn’t Everything; it’s the Only Thing” – T. Marzetta

We need to know where the point the beam!

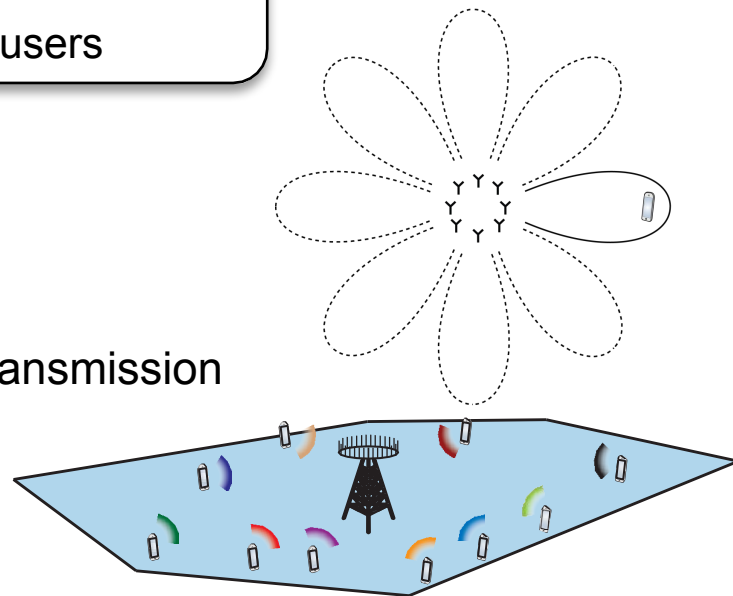
- Give strong received signal
- Limit interference between users

- **Conventional approach:** Grid-of-beams

- Try 8 angular beams, user reports the best one
- Good: Simple, works with both TDD and FDD
- Bad: Only works in line-of-sight and single-user transmission

- **Massive MIMO:** Uplink estimation

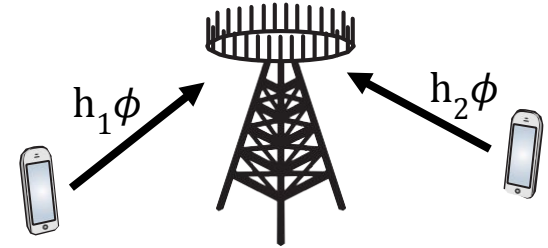
- User sends pilot signal, BS estimates channel
- Good: Works in any propagation scenario, scalable for large arrays
- Bad: Only works in TDD, where uplink estimates useful for downlink



Pilot Contamination

- Pilot Length τ_p Limited by Channel Coherence
 - Only τ_p orthogonal pilots
 - Some users reuse pilot ϕ

- Received pilot: $y = h_1\phi + h_2\phi + w$
 - Estimate: $\hat{h}_1 = y/\phi = h_1 + h_2 + w/\phi$



Channels: $h_k \sim CN(0, I_M)$

Noise: $w \sim CN(0, I_M)$

Received signal with maximum ratio filter:

Asymptotically:
Signal and interference
from pilot-sharing users
remain!

$$\begin{aligned} \text{Signal remains: } v_1^H h_1 &= \frac{1}{M} (\|h_1/\phi\|^2 + \cancel{h_1^H h_2} + \cancel{w^H h_1}) \xrightarrow{M \rightarrow \infty} 1 \\ \text{Interference remains: } v_1^H h_2 &= \frac{1}{M} (\cancel{h_1^H h_2} + \cancel{\|h_2\|^2} + \cancel{w^H h_2}) \xrightarrow{M \rightarrow \infty} 1 \end{aligned}$$

Will Pilot Contamination Limit Performance?

With i.i.d. Rayleigh fading

- Yes! Marzetta showed it in 2010
- Has encouraged a lot of research

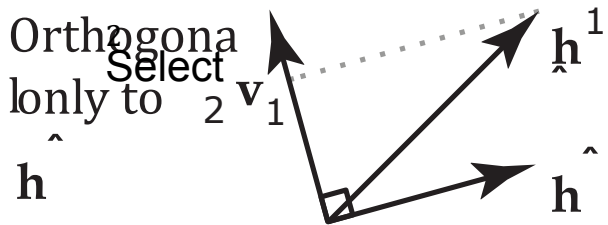
In general: No!

Natural power variations change everything!

Utilize user's spatial characteristics:

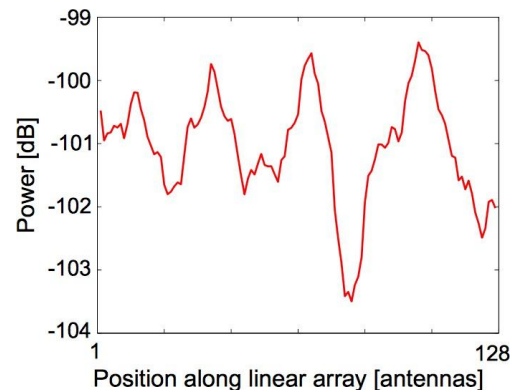
$$\text{Receive } y = h_1 \phi + h_2 \phi + w$$

Obtain linearly independent $\hat{h}_1$ and $\hat{h}_2$



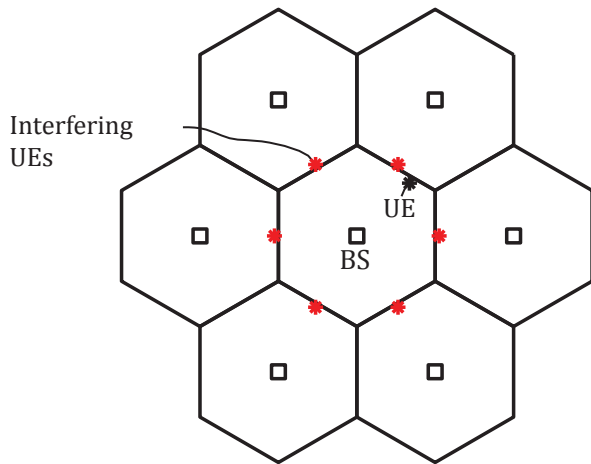
$$v_1^H \hat{h}_2 = 0$$

Recall:
Power variations over
an array

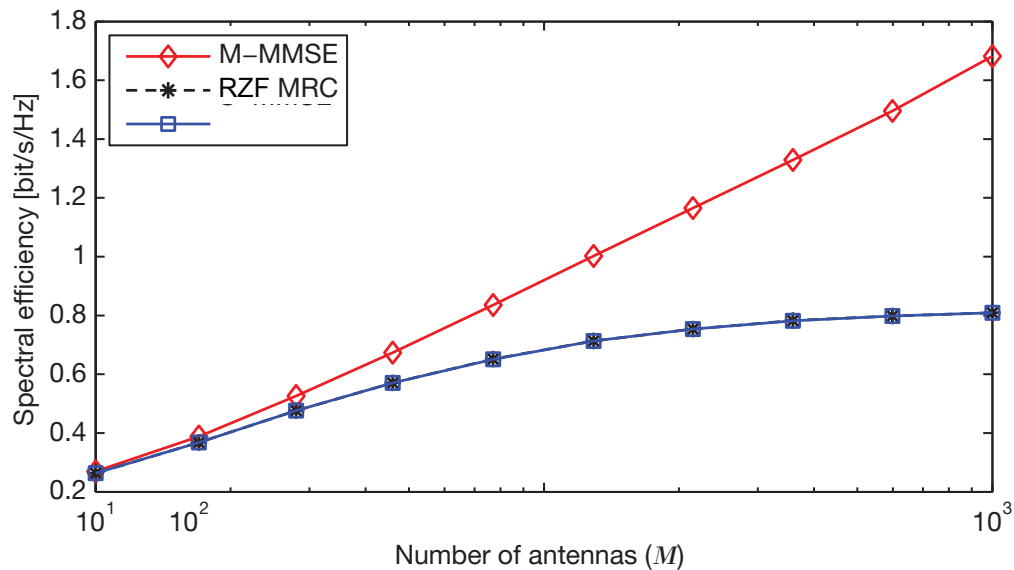


E. Björnson, J. Hoydis, L. Sanguinetti, "Massive MIMO has Unlimited Capacity", *Transactions on Wireless Communications*, to appear

How Big Can We Make Massive MIMO?



Everyone uses the same pilot!



Spectral efficiency grows logarithmically with M
If utilizing spatial characteristics using M-MMSE filter

Implications:

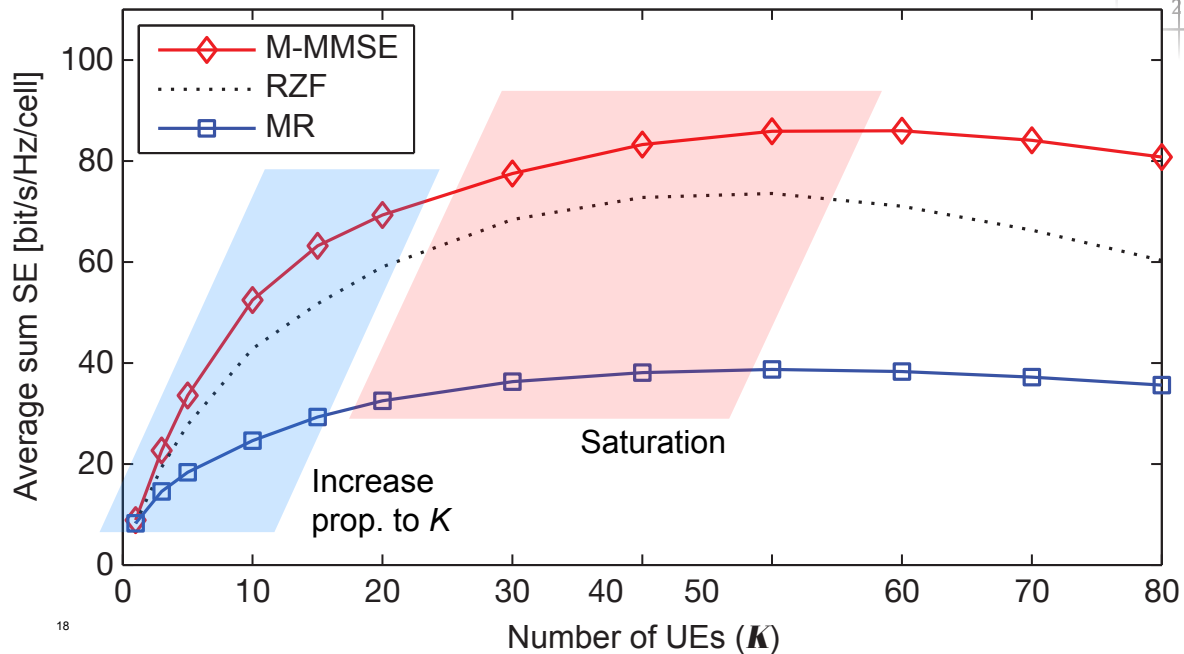
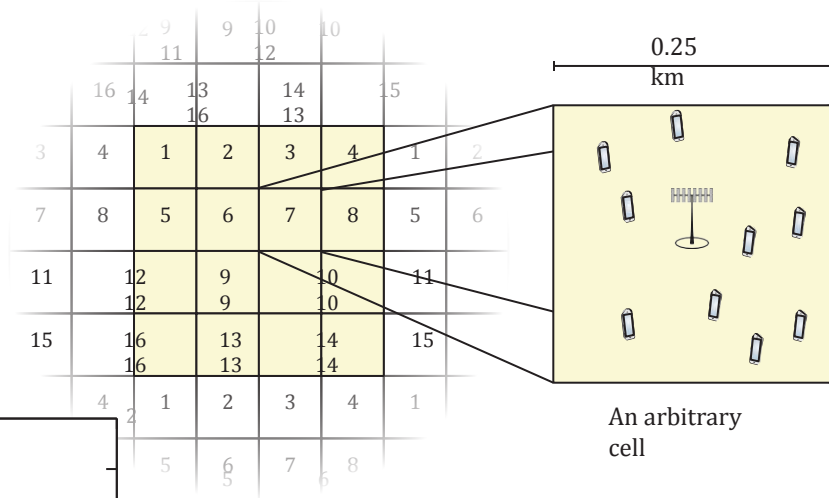
- Can always handle more traffic by adding more antennas
- But diminishing returns when increasing M

Practical Implications

Simulation

100 antennas per base station

Channel is fixed for 200 symbols

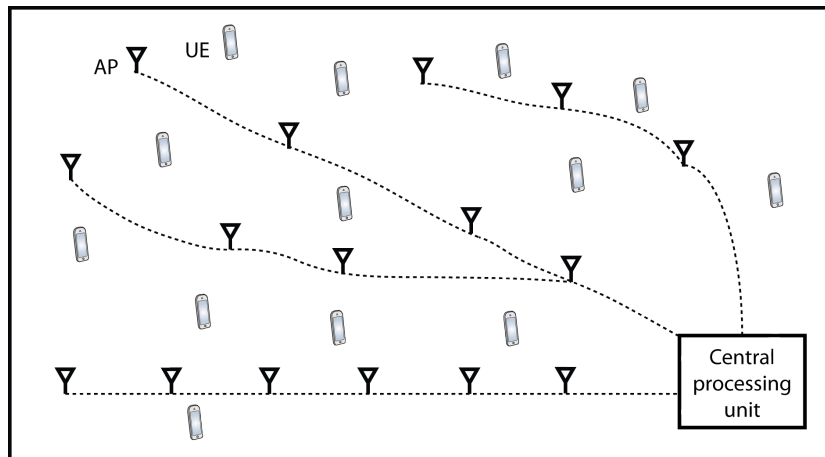
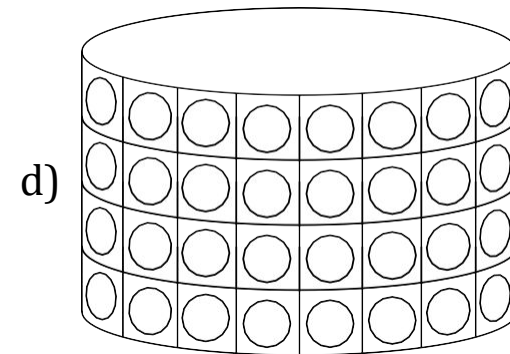
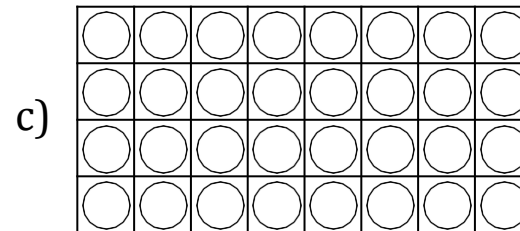
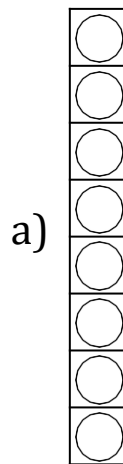


**Identify maximum K
in your application**

Select M to make the
sum spectral efficiency
right for this K

How to Deploy Many Antennas?

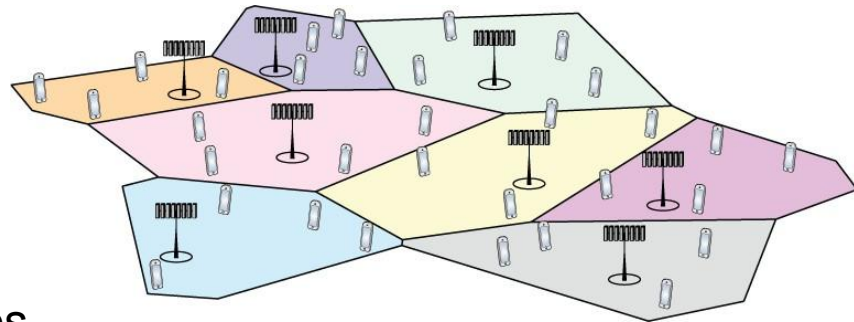
Array geometries:
1D, 2D with sectors
3D without sectors



- Distributed deployment**
1. Multiple arrays per cell
 2. Cell-free approach

Summary: How Big Can Massive MIMO Get?

- Massive MIMO
 - Delivers extreme spectral efficiency
- No Theoretical Capacity Limit
 - Grows towards infinity as M increases
 - Increase M when higher spectral efficiency is needed
 - Pilot contamination can be dealt with
- Practical Limits
 - 100-antenna arrays not larger than in 4G
 - Increasing cost of deploying base stations
 - Distributed deployment in the future



Learn More: Blog and Book

- Many Online Resources
 - Info Point (<https://massivemimo.eu>): Library with research papers
 - LinkedIn Group: “Massive MIMO for 5G”



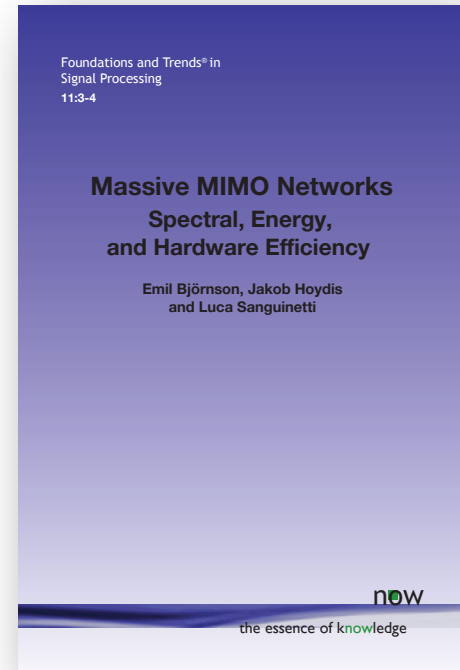
www.massive-mimo.net

Upcoming book:

Emil Björnson, Jakob Hoydis
and Luca Sanguinetti (2017),
**“Massive MIMO Networks:
Spectral, Energy, and
Hardware Efficiency”**

500 pages, Matlab code

www.massivemimobook.com



Thank you!

Slides, papers, and code available online:

<http://www.ebjornson.com/research>

<http://www.massivemimobook.com>